THE DEGREE OF THE BASE FIELD IS ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the identification of the cycle class of the base field with one, and the computation of the degree through multiplicativity for the square of the base field.

Editorial note. The label lemma-trace-base names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-trace-base –start 4993 –end 5010 –local-id 28169 returned RESULT=OK. Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The cycle class of the base field is one by construction. The construction of the cycle class map, which sends the fundamental class of a scheme to the unit. Identifies the cycle class of the spectrum of the base field with one.
- Multiplicativity of the degree map for products. Same chapter, the lemma that the degree map of a product is the product of the degree maps. Relates the degree map of the square of the base field to the square of the degree map of the base field.

Source reference. Original reference 0FIL: sources/stacks-weil/pinned-original-source.tex, source lines 4947–4947. Prerequisite pointer invoked at proof line(s) 5009; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07XJ: sources/stacks-weil/pinned-original-source.tex, source lines 2568–2568. Prerequisite pointer invoked at proof line(s) 4996; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assume given data (D0), (D1), and (D2') satisfying axioms (A1) – (A7). Then axiom (C)(d) of Section 07XJ holds.*

Proof. We have $\gamma([\mathrm{Spec}(k)]) = 1 \in H^*(\mathrm{Spec}(k))$ by construction. Since

$$H^0(\mathrm{Spec}(k)) = F, \quad H^0(\mathrm{Spec}(k) \times \mathrm{Spec}(k)) = H^0(\mathrm{Spec}(k)) \otimes_F H^0(\mathrm{Spec}(k))$$

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the map $\int_{\mathrm{Spec}(k)} = \lambda$ of axiom (A6) must send 1 to 1 because we have seen that $\int_{\mathrm{Spec}(k)\times\mathrm{Spec}(k)} = \int_{\mathrm{Spec}(k)} \int_{\mathrm{Spec}(k)}$ in Lemma 0FIL. $\square$